

Module 14

Summarizing Cloud Concepts

Learning Objectives

- Explain datacenter and storage network architecture.
- Summarize cloud concepts.
- Summarize the use of software, coding, and zero trust in modern network environments.

Lesson 14.1

Datacenter and Storage Networks

Datacenter Network Design

Components of Modern Datacenter Network Design

- Virtualization: Agile and scalable networks
- Overlay Networks: Secure, flexible server-to-server traffic
- Zero Trust Architecture: Authentication/authorization for each request
- High Availability features: Networking, power, climate, and access control for reliability

Understanding Traffic Flows: North-South vs. East-West

- North-South Traffic: External client access to datacenter resources
- East-West Traffic: Server-to-server communication within datacenter
- Security and performance challenges: Need for advanced security and efficient traffic management

Spine and Leaf Topology

Spine and Leaf Topology

- Two-layer network topology for optimized data center traffic flow

Structure

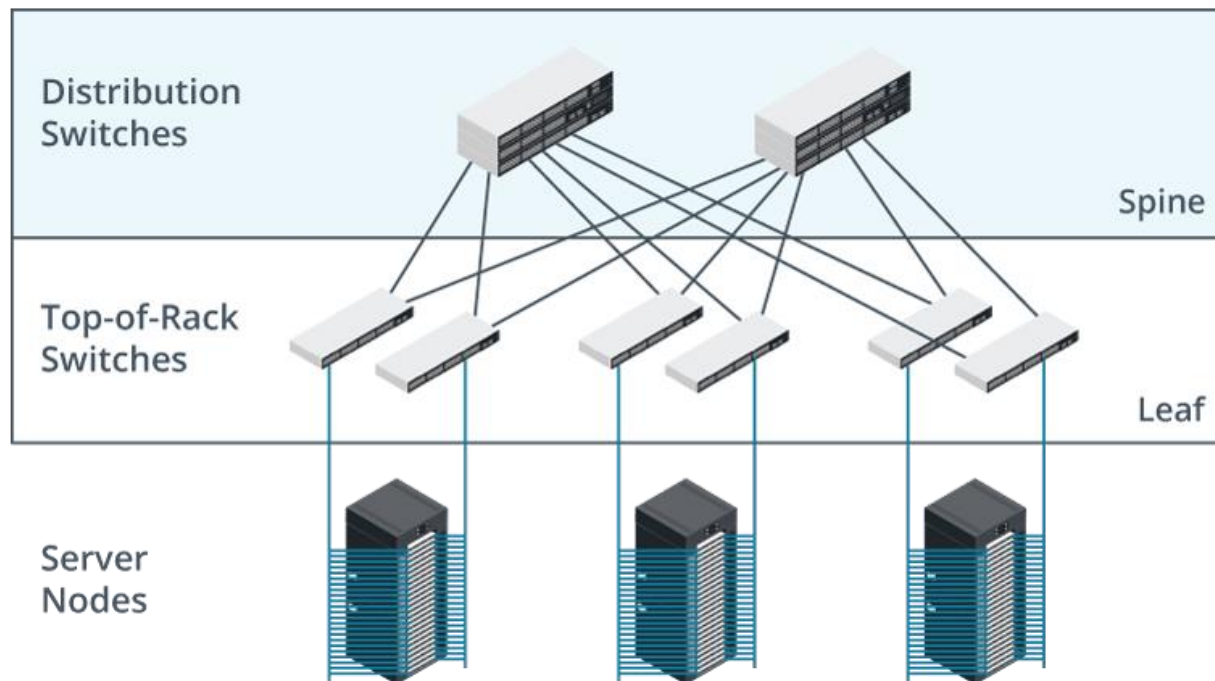
Spine Layer

- Backbone containing top-tier distribution switches
- Spine switches interconnected with all leaf switches but not with each other

Leaf Layer

- Includes access switches that connect to every spine switch in a full mesh topology
- Facilitates direct paths from devices to the network backbone without inter-leaf switch connections

Spine and Leaf Topology Example



Benefits of Spine and Leaf Topology

Predictable Latency

Single hop from any server to the backbone

Ensures consistent network performance

Load Balancing and Failover

Multiple redundant paths between leaf and spine

Allow for effective load distribution and reliability

Loop-Free Design

Non-interconnected spine and use of ECMP on leaf switches

Eliminates need for Spanning Tree Protocol

Scalability and Redundancy

Easy to scale by adding more spine or leaf nodes without topology changes

Servers connect to multiple leaves for redundancy

High-Speed Connectivity

Top-of-Rack (ToR) switches at the leaf layer provide high bandwidth connections to servers

Support modern data-intensive applications

Storage Area Network (SAN)

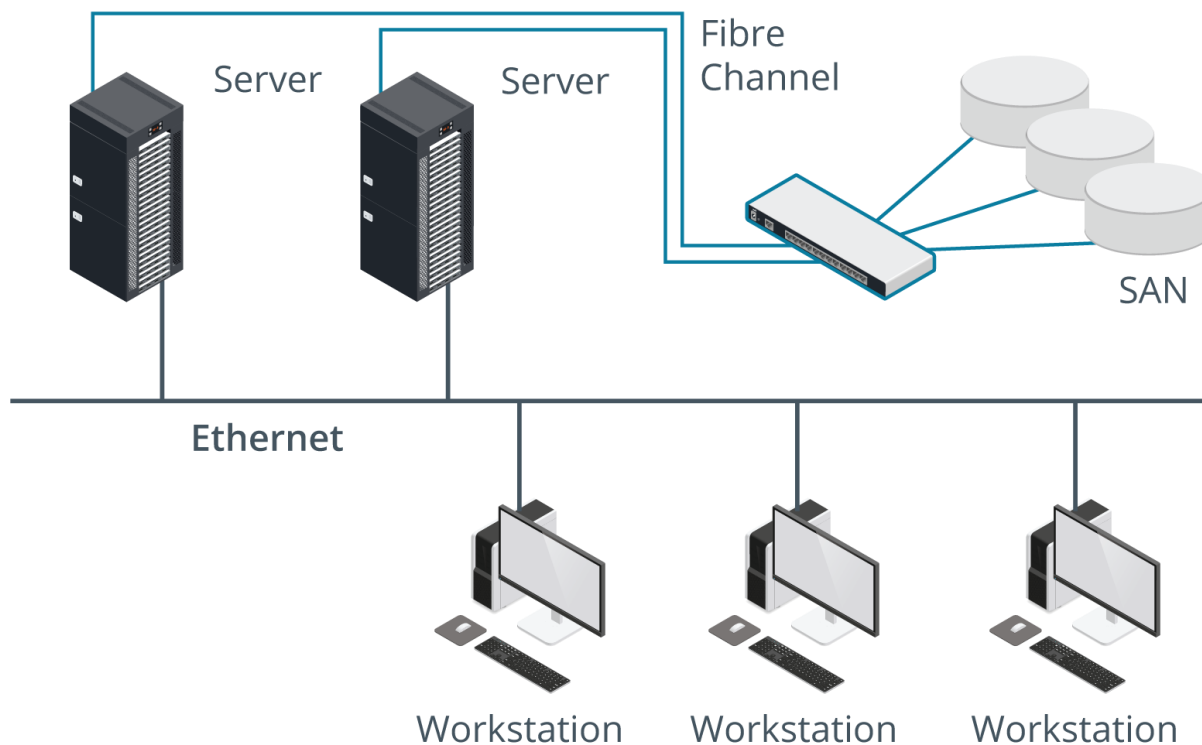
Storage Area Network (SAN)

- A network dedicated to providing access to consolidated, block-level data storage
- Separates storage resources from the local network

Components of a SAN

- Storage devices (RAID arrays, tape libraries)
- SAN Fabric (Switches and directors that connect servers with storage devices)
- SAN Software (Management and configuration tools)

SAN Example



Storage Area Network (SAN)

Benefits of a SAN

- High Availability
- Scalability
- Minimized latency and maximized throughput

Use Cases

- Database management
- Business continuity
- Cloud storage

Fibre Channel (FC)

Defined

High-speed data transfer protocol

Backbone for SANs

British spelling “fibre” to distinguish from fiber optic cables

Components

Initiators (servers with a host bus adapter)

Targets (storage device ports)

FC switches (facilitate data transfer)

Advantages

Performance

Flexibility

Speed

Quality of Service (QoS) mechanisms

Activity: Pop Quiz

What is the difference between
North-South traffic flow and
East-West traffic flow?



What is the difference between **north-south** traffic flow and **east-west** traffic flow?

- a) Direction: Flows in and out of a network or data center.
- b) Direction: Flows within a network or data center, primarily between servers and storage devices.
- c) Destination: Often directed towards the internet, cloud platforms, or centralized resources outside the local network.
- d) Destination: Directed towards other servers or storage devices within the local network.

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Lesson 14.2

Cloud Concepts

Scalability in Cloud Computing

Definition	Types	Benefits	Examples
• Capability of a cloud computing system to handle growing amount of work • Cloud system grows with your needs: Handles increasing workloads	• Horizontal Scaling (Scaling Out/In) • Vertical Scaling (Scaling Up/Down)	• Flexibility • Performance	• Scale servers for anticipated surge in web traffic (product launch) • Upgrading server specifications to support more intensive compute tasks

Elasticity in Cloud Computing

Definition

- Ability of a system to automatically adjust and allocate computing resources on demand
- Ensures optimal performance despite fluctuations in usage

Key Features

- Automatic Scaling
- Cost-Effectiveness (pay what you use)
- Real-Time Responsiveness

Examples

- Auto-scaling web application servers during a traffic spike
- Decreasing allocated resources overnight when demand is lower to reduce costs

Cloud Deployment Models

Model	Description	Pros	Cons
Public Cloud	Shared resources over the Internet by CSPs	Cost-effective, pay-as-you-go financing, subscriptions	Higher risks in performance and security
Hosted Private Cloud	Exclusively used by one organization, hosted by a third party	More secure, better performance	More expensive
Private Cloud	Owned and operated by the organization	Greater control over privacy and security	Best for services requiring strict access control
Hybrid Cloud	Mixes public and private cloud solutions	Flexibility to scale; balances security and cost	Complexity in managing multiple environments

Cloud Service Models

Infrastructure as a Service (IaaS)

- Provides virtualized computing resources over the internet
- Examples: AWS EC2, Microsoft Azure Virtual Machines, OpenStack

Software as a Service (SaaS)

- Delivers software applications over the internet (subscription basis)
- Examples: Microsoft Office 365, Salesforce, Google Workspace

Platform as a Service (PaaS)

- Hardware and software tools over the internet, typically for application development
- Examples: Oracle Database, Microsoft Azure SQL Database, Google App Engine

Advantages of Cloud Service Models

Scalability and Flexibility

- Scale resources up or down based on demand without the need for physical hardware changes

Cost Efficiency

- Pay only for the computing resources you use

Speed and Efficiency

- Deploy and manage applications faster without the need to set up and maintain physical servers

Accessibility and Collaboration

- Access your applications and data from anywhere

Content Delivery Networks (CDNs)

What is a CDN?

Caching solution for high-traffic websites and apps

Enhances availability, performance, and scalability

How Does It Work?

Distributes servers and media files near the consumer for faster access

Content can include web apps, websites, scripts, downloads, and more

Benefits of CDNs

Distributed across multiple, geographically dispersed data centers

Quick data access by placing content closer to the consumer

Easily handles spikes in traffic and data demands

Mechanism

Content is delivered via multiple Internet backbones, ensuring reliability

CDN servers replicate content among themselves to stay updated

Lesson 14.3

Cloud Networking

Cloud Instances

Types of Cloud Instances

- **Virtual Machines (VMs):** Fully-fledged servers with emulated hardware, can run an OS and multiple applications
- **Containers:** Isolated environments for running applications (lightweight & portable)
- **Virtual Appliances:** Pre-configured VMs/containers (emulating hardware functions)

Advantages of Cloud Instances

- **Flexibility and Scalability:** Easily resized and scaled; no physical hardware changes
- **Cost-Effectiveness:** Scale costs with resource usage
- **Rapid Deployment:** Fast provisioning; minutes to deploy & configure instances

Comparing Cloud Instances

Virtual Machines (VMs)

Run multiple OSs on a single hardware platform

Complete isolation from the host system (secure and stable operations)

Higher resource consumption due to emulation of hardware

Virtual Appliances

Pre-built software packages for tasks like firewalls, ready to run

Optimized to use resources more efficiently than general-purpose VMs for specific tasks

Can be easily deployed across diverse cloud environments

Containers

Shares the host system's kernel, reducing overhead significantly

Easier to scale and manage due to their small footprint

Designed to run a single application or service for performance and portability

Virtual Private Clouds

Virtual Private Clouds (VPCs)

- Serve as isolated environments within a public cloud
- Enable organizations to run their applications securely

Key Features

- Each VPC is isolated from other tenants in the cloud
- Tenants can configure their virtual network
- Offer multiple layers of security controls, including security groups and ACLs

Use Cases

- Application Hosting
- Hybrid Cloud Environments
- Customization and Flexibility
- Networking Across Clouds

Cloud Gateways

Public vs. Private Subnets

- Public Subnet
 - Requires attaching an Internet gateway and configuring it as the default route
- Private Subnet
 - Remains private without a default Internet gateway route, even if an Internet gateway is attached

Internet Gateway

Internet Gateways

- Enables two-way Internet communications
- Instances must have a public IP

NAT Gateway

- Allows outbound Internet access without inbound
- No public IP needed for VM

VPN Options

- Securely connect VPCs through software-based VPNs or CSP-managed VPN services

Addressing

- Public IPs can be from a personal pool or CSP services (Amazon's Elastic IP)

Cloud Connectivity Options

VPN (Virtual Private Network)

- Cost-effective, easy to implement
- Best for distributed networks
- Challenges: potential latency/bandwidth issues

Direct Connect/Colocation

- High bandwidth, low latency
- Ideal for centralized operations
- Requires physical infrastructure within or connected to a datacenter

Transit Gateways

- Simplifies network architecture
- Enables multi-VPC and VPC-to-on-premises connections
- Offers scalability and ease of management

Cloud Firewall Security

Cloud Firewall Security

- Acts as a barrier between cloud resources/external threats
- Enforces security rules to allow or block traffic

Functions

- Segmentation for performance and compliance
- Filters traffic based on OSI model layers

OSI Layers Explained

- Layer 3 (Network): IP addresses and TCP/UDP ports filtering
- Layer 4 (Transport): Stateful inspection of connections
- Layer 7 (Application): Parsing protocol headers for content-based filtering

Implementing Cloud Firewall Solutions

Implementation Options

- Host-based software on instances
- Service at the virtualization layer

Considerations

- Software-based solutions consume resources
- Management complexity with multiple instances

Benefits of Cloud Firewalls

- Scalable and flexible security management
- Facilitates compliance with data protection laws

Cost vs. Efficiency

- Analysis of transaction costs for cloud firewalls
- When to consider third-party firewall services

Security Groups

Definition

- A security group acts as a virtual firewall for instances to control inbound and outbound traffic

Functionality

- Operates at the instance level
- Provides stateful filtering
- Allows specifying allow rules, but all unmatched traffic is dropped

Key Points

- Multiple security groups can be assigned to a single instance
- Customization allows for specific ports and IP ranges to be defined for access
- Restricting traffic based on protocol, port numbers, and source/destination IP addresses

Security Lists

Functionality

- Operates at the subnet level rather than the instance level
- Provides stateless filtering
- Allows setting both allow and deny rules to finely control the traffic flow

Key Points

- A single security list can affect multiple instances within the same subnet
- Ideal for implementing broad network policies that apply uniformly to all instances in a subnet

Activity: Matching

Fully-fledged servers with emulated hardware, capable of running an operating system and multiple applications

Lightweight, used for running single applications or processes with minimal overhead

Pre-configured; often used for emulating specific hardware functions like routers or firewalls

Containers

Virtual Machines

Virtual Appliances

Virtual Machines

Fully-fledged servers with emulated hardware, capable of running an operating system and multiple applications

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Virtual Appliances

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Lesson 14.4

Modern Network Environments

Infrastructure as a Code

Infrastructure as a Code (IaC)

- Management of IT infrastructure through machine-readable definition files
- Uses code to manage and provision the IT infrastructure automatically and safely
- Enables quick, consistent, and cost-effective deployment of applications across various environments

Benefits

- Consistency and efficiency
- Speed and scalability
- Version control and collaboration

Tools for Infrastructure as a Code

Terraform

- Allows for declarative coding to provision and manage any cloud, infrastructure, or service

Ansible

- Uses a declarative language to automate deployment, configuration, and orchestration tasks

Chef

- Uses recipes and cookbooks to apply configurations across environments

Puppet

- Uses a declarative approach to automate infrastructure management

AWS Cloud Formation

- Allows you to use programming languages or a simple text file to model and provision all the resources needed for your applications

Use Cases for Infrastructure as a Code

Automating Deployment

- Rapidly deploy VMs and containers to various cloud environments with minimal manual intervention

Consistency & Speed

- Use master images or templates for consistent and fast provisioning of infrastructure

Efficient Upgrades

- Streamline the upgrade process of OS and software versions

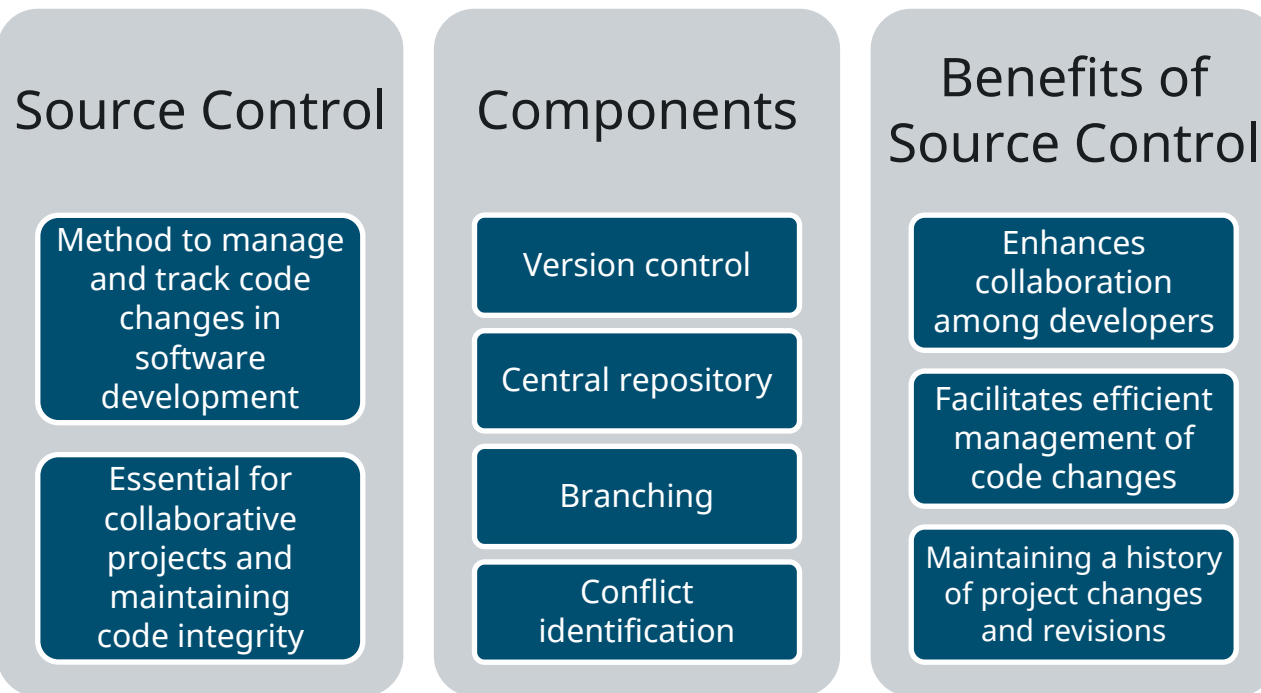
Dynamic Inventory Management

- Keep track of cloud instances through dynamic inventories

Reusable Tasks & Scalability

- Create modular, reusable blocks of code for various tasks

Source Control



Software-Defined Networking (SDN)

Definition

- Simplifies network management by separating control and forwarding functions
- Allows direct programmability of network control and abstraction of the infrastructure for applications and services

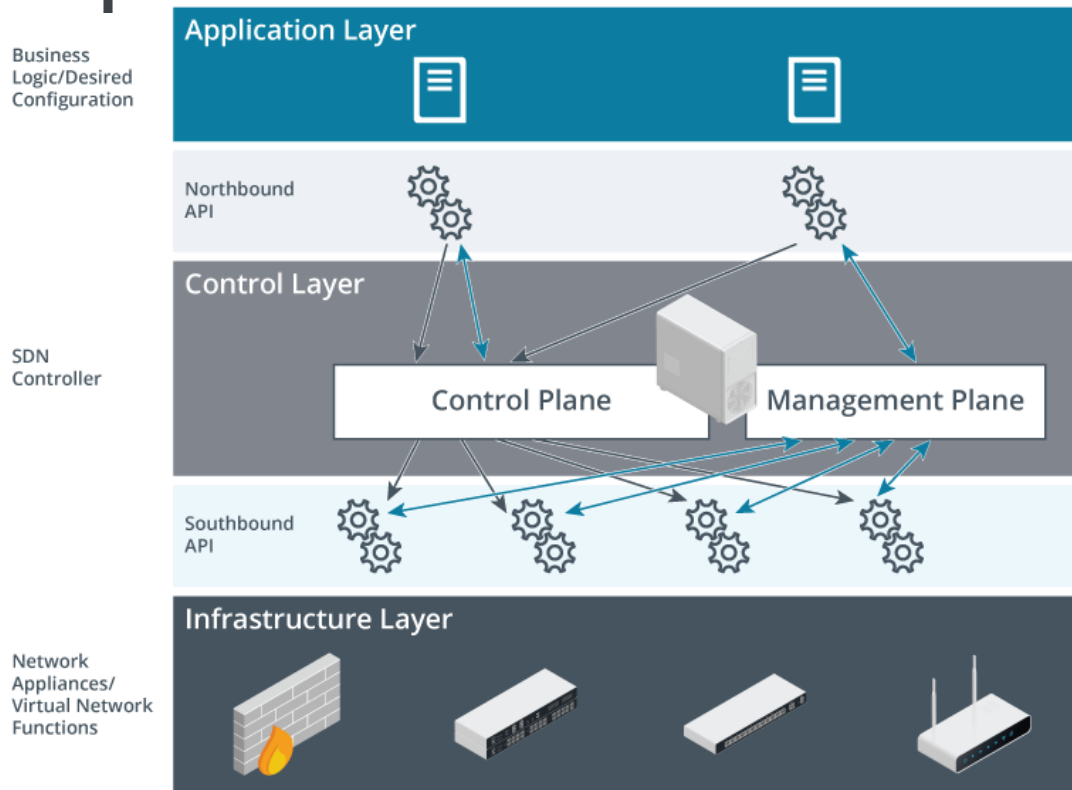
Components

- SDN Controller (Brain of the network)
- Southbound APIs (Communication between controller and devices)
- Northbound APIs (Interface between controller and the business applications)

Benefits

- Enhanced network agility and flexibility
- Simplified network management
- Improved network efficiency and performance

SDN Example



SDN Use Cases

Network Virtualization

- Creating multiple virtual networks with distinct security policies and functions over a single physical infrastructure

Cloud Computing

- Dynamically adjusting network resources to meet the demands of cloud services, improving scalability and reducing operational costs

Data Center Management

- Automated provisioning and management of network resources, enhancing data flow efficiency and reducing latency

IoT (Internet of Things):

- Managing the massive and diverse traffic from IoT devices for reliable connectivity and security at scale

Overlay Networks

Overlay Network

- Virtual network built on top of an existing physical network
- Allows for logical links between nodes or different networks

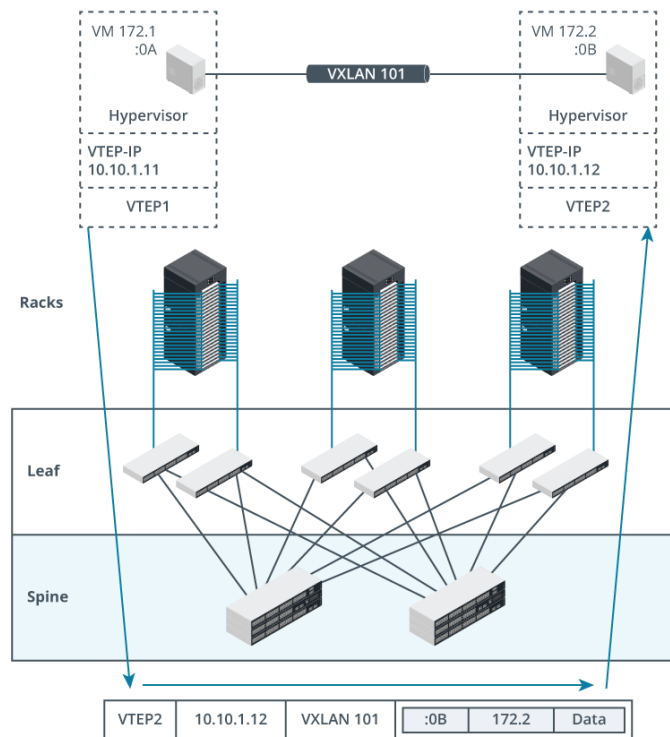
Examples

- Virtual Private Networks (VPNs)
- Networks using encapsulation protocols SDN

Components

- Uses encapsulation to create virtual tunnels
- Virtual Extensible LAN (VXLAN) for data center implementations
- Facilitates network segmentation and isolation

Overlay Network Example



Software-Defined WAN

Software-Defined WAN

- Innovative approach to designing and deploying enterprise WAN
- Enhanced connectivity between branch offices, datacenters, and the cloud
- Traffic flow optimization: automation, central management, and secure IPsec tunnels

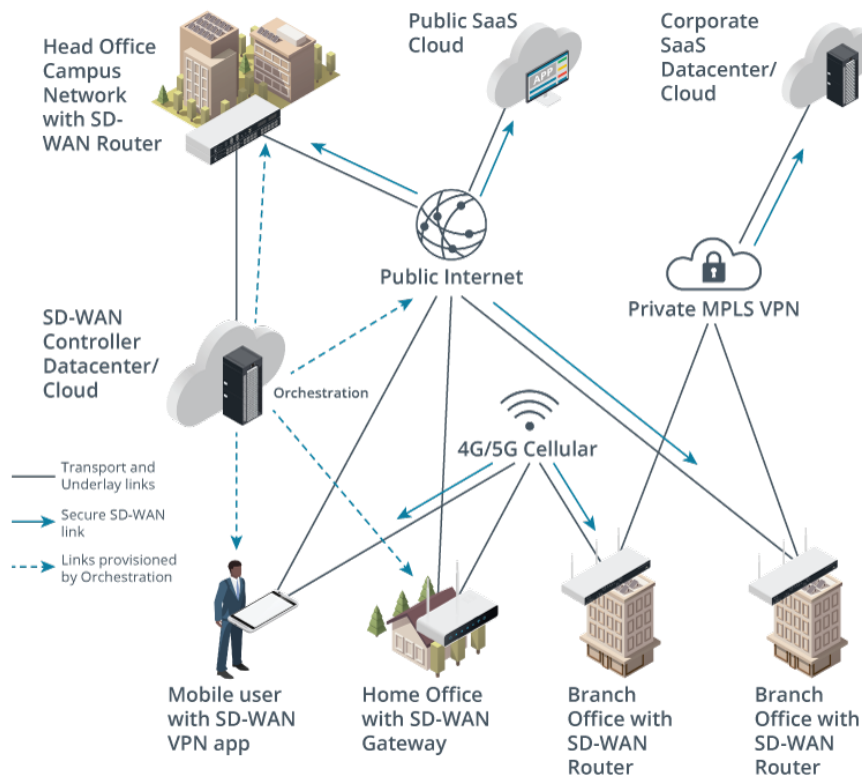
Components

- SD-WAN Controller & Management Software
- SD-WAN Devices

Benefits

- Improved Performance
- Increased Security
- Cost Savings

Software-Defined WAN Example



Zero Trust Architecture – Core Principles

Assume No Trust

Operate under the assumption that no entity (user or device) is trusted by default (inside/ outside network)

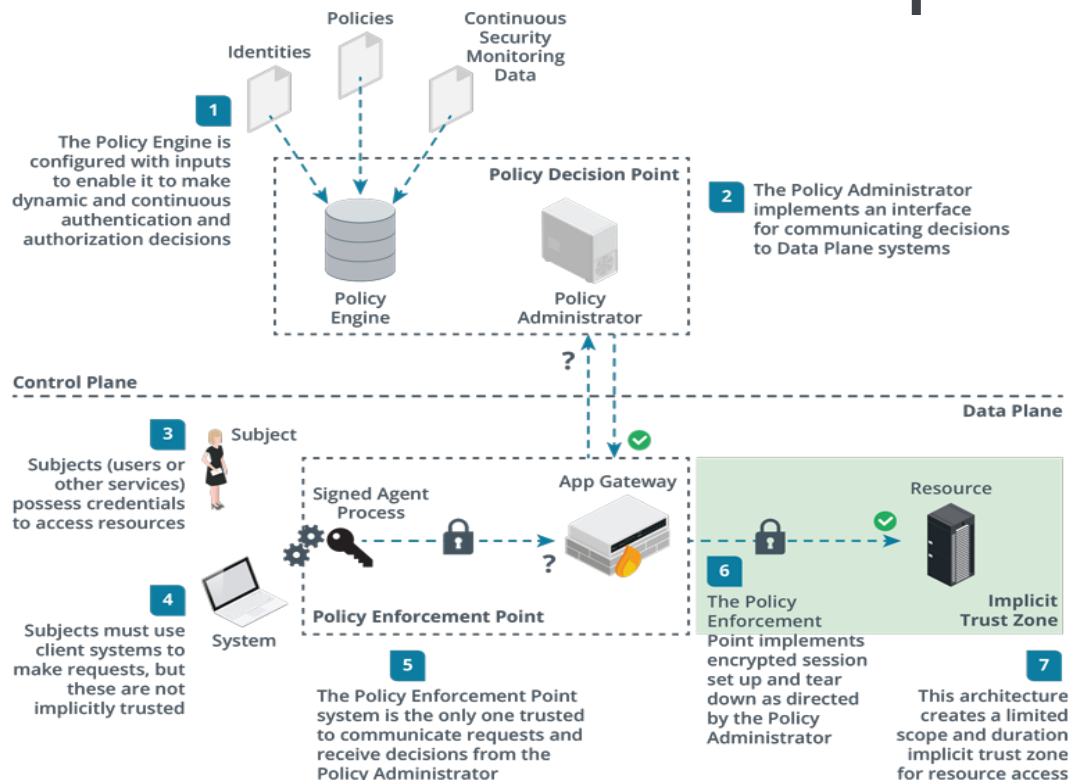
Continuous Verification

Implement constant authentication and authorization of every user/device for every session/ transaction

Least Privilege Access

Grant users and devices minimum levels of access (permissions) needed to accomplish tasks to minimize potential attack vectors

Zero Trust Architecture Example



Implementing Zero Trust Architecture

Policy-driven Authentication and Authorization

Use centralized
policy decision
points for dynamic
security
assessments based
on user behavior,
device health, and
network context

Separation of Control and Data Planes

Maintain distinct
mechanisms for
policy definition
and enforcement
to reduce risks and
ensure granular
access control

Implicit Trust Zones and Microsegmentation

Establish secure
pathways for data
based on stringent
policy enforcement

Secure Access Service Edge (SASE)

Secure Access Service Edge (SASE)

- Cloud-based architecture combining network and security functions with widespread global reach
- Integrates SD-WAN technology and Security Service Edge (SSE)

Benefits of SASE

- Simplifies WAN architecture
- Enhances security posture
- Reduces complexity and overhead
- Improves performance and user experience

Components

- SD-WAN Technology
- Secure Service Edge (SSE)

Secure Service Edge (SSE)

Functions of SSE

- Zero Trust Architecture
- Cloud Access Security Broker (CASB)

Functions of CASB

- Manages and enforces security policies
- Scans for malware
- Monitors and audits activity
- Mitigates data exfiltration

CASB Implementation Methods

- Forward Proxy: Acts as a gatekeeper, monitoring and controlling outbound client traffic
- Reverse Proxy: Cloud gatekeeper, enforces access policies for services
- API: Direct connections for cloud service access & control

Summary

- **Cloud Strategy:** Evaluate cloud apps & services; choose deployment model (public, private, hybrid) & service type (IaaS, SaaS, PaaS, DaaS) for optimal scalability & elasticity.
- **Cloud App Development:** Leverage IaC for automation and leverage SDN benefits.
- **Private Cloud/Datacenter:** Virtualization & SAN for enhanced elasticity & scalability. Integrate with SDN & NFV.
- **Public Cloud Security:** Define Cloud Responsibility Matrix and conduct regular risk assessments & security audits.

Thank you!

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